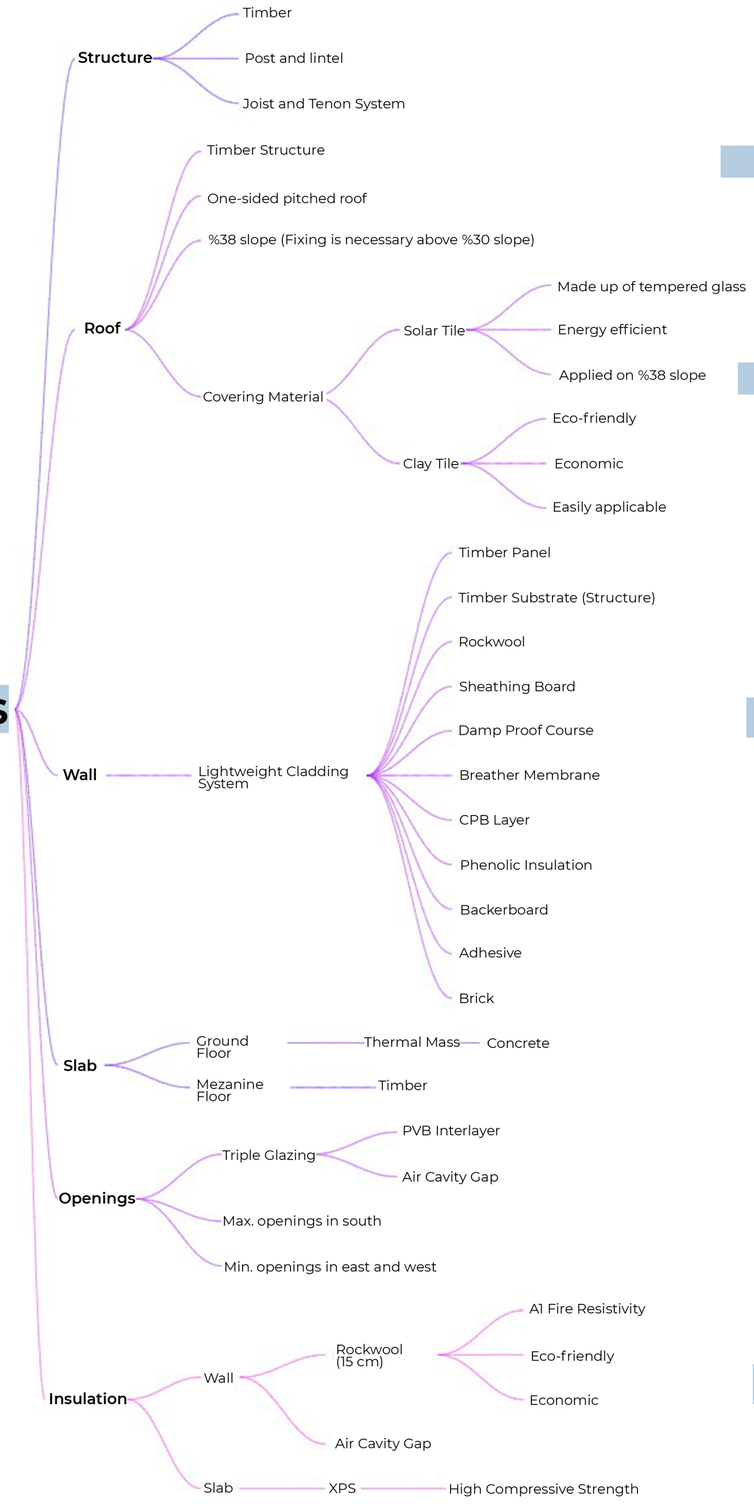
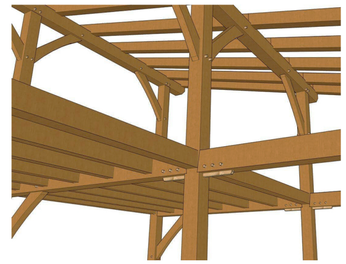
ARCH 351 - The Term Project

Phase 1 Cihan Ay Cihat İlkbahar Sena Nur Cabadağ

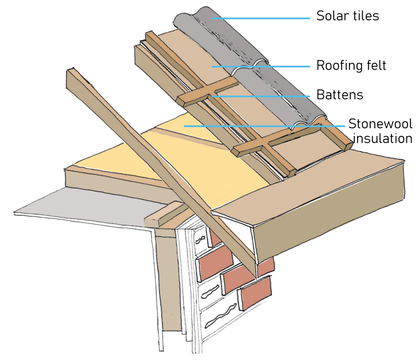
Passivhaus
in Ankara



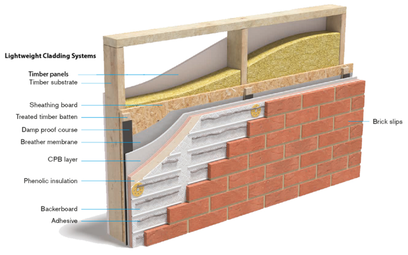
The choosen site is located at **Ankara - Yüzüncüyıl, which has cold-dry climatic conditions and high precipitation.** The proposed house is designed to be ***Passivhaus with sustainable construction techniques and materials.*** The structural system is based on timber framing. The components from top to the bottom are;

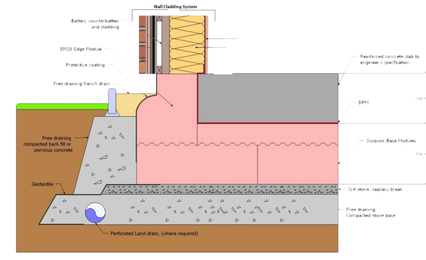


1-**One sided pitched cold roof based on timber structure covered with the hybrid use of solar and clay tiles.** About %38 slope is applied to the roof due to high snow load in Ankara and slope is oriented facing south to benefit solar tiles effectively. Thermal insulation is provided by rockwool with cold roof system.



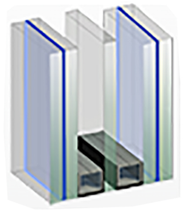
2- **Composition of timber and brick wall with high thermal insulation** between them (rockwool in min. 15 cm thickness). Material selection is done due to sustainability reasons. Damp proof course and breather membrane added as interlayers. For fire resistivity, Phenoic Insulation is used.

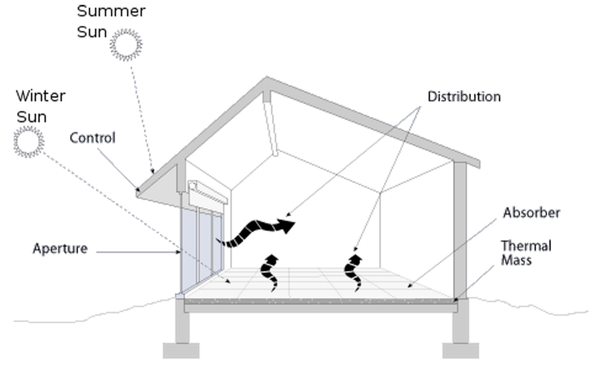




3- **Concrete as the thermal mass for ground floor slab and timber for the mezanine floor slab.** High thermal mass is needed for preventing heat loss.

4-Maximum number of openings facing south to get heat in winter and overhangs for preventing excessive heat gain in summer. **Triple glazing composition with PVB interlayer** is chosen to prevent heat escape. The required u value is 2,4 W/(m².K)





5- **Thermal Insulation with rockwool in walls and roof, XPS for ground** due to its high compressive strength.

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